

● MEDIUM

● ADVANCED MATH

Advanced Math

09

10 questions · ~15 min

Q1

ADVANCED MATH

Which expression is equivalent to $\sqrt[7]{x^9y^9}$, where x and y are positive?

- A. $xy^{\frac{7}{9}}$
- B. $xy^{\frac{9}{7}}$
- C. xy^{16}
- D. xy^{63}

x	hx
0	1.23
2	1.54
4	1.94

The table shows the exponential relationship between the number of years, x , since Hana started training in pole vault, and the estimated height hx , in meters, of her best pole vault for that year. Which of the following functions best represents this relationship, where $x \leq 4$?

- A. $hx = 1.120.23^x$
- B. $hx = 1.121.23^x$
- C. $hx = 1.230.12^x$
- D. $hx = 1.231.12^x$

Q3

ADVANCED MATH

$$w^2 + 12w - 40 = 0$$

Which of the following is a solution to the given equation?

- A. $6 - 2\sqrt{19}$
- B. $2\sqrt{19}$
- C. $\sqrt{19}$
- D. $-6 + 2\sqrt{19}$

Q4

ADVANCED MATH

If $p = 3x + 4$ and $v = x + 5$, which of the following is equivalent to $pv - 2p + v$?

- A. $3x^2 + 12x + 7$
- B. $3x^2 + 14x + 17$
- C. $3x^2 + 19x + 20$
- D. $3x^2 + 26x + 33$

$$h(t) = -16t^2 + 110t + 72$$

The function above models the height h , in feet, of an object above ground t seconds after being launched straight up in the air. What does the number 72 represent in the function?

- A. The initial height, in feet, of the object
- B. The maximum height, in feet, of the object
- C. The initial speed, in feet per second, of the object
- D. The maximum speed, in feet per second, of the object

Q6

ADVANCED MATH

$$6x^2 + 5x - 7 = 0$$

What are the solutions to the given equation?

- A. $\frac{-5 \pm \sqrt{25 + 168}}{12}$
- B. $\frac{-6 \pm \sqrt{25 + 168}}{12}$
- C. $\frac{-5 \pm \sqrt{36 - 168}}{12}$
- D. $\frac{-6 \pm \sqrt{36 - 168}}{12}$

Q7

ADVANCED MATH

Which of the following expressions is equivalent to $8x^{10} - 8x^9 + 88x$?

- A. $x7x^{10} - 7x^9 + 87x$
- B. $x8^{10} - 8^9 + 88$
- C. $8xx^{10} - x^9 + 11x$
- D. $8xx^9 - x^8 + 11$

Q8

ADVANCED MATH

The product of a positive number x and the number that is 8 more than x is 180. What is the value of x ?

- A. 5
- B. 10
- C. 18
- D. 36

Q9

ADVANCED MATH

$$\frac{1}{7b} = \frac{11x}{y}$$

The given equation relates the positive numbers b , x , and y . Which equation correctly expresses x in terms of b and y ?

- A. $x = \frac{7by}{11}$
- B. $x = y - 77b$
- C. $x = \frac{y}{77b}$
- D. $x = 77by$

$$\left(\frac{1}{2}x + \frac{3}{2}\right)\left(\frac{3}{2}x + \frac{1}{2}\right)$$

The expression above is equivalent to $ax^2 + bx + c$, where a, b, and c are constants.

What is the value of b?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.